

Epistasis limits stepwise strain engineering: a complete three-factor test in fatty acid production (13/15 words)

Michael Volk^{1,2,3*} and Huimin Zhao^{1,2,3*}

¹DOE Center for Advanced Bioenergy and Bioproducts Innovation, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA.

²Department of Chemical and Biomolecular Engineering, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA.

³Carl R. Woese Institute for Genomic Biology, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA.

*Corresponding author(s). E-mail(s): mjvolk3@illinois.edu; zhao5@illinois.edu;

Abstract (≤ 150 words)

Strain engineering proceeds one edit at a time, keeping what improves and discarding what does not. The procedure assumes that an improving combination can be reached through improving intermediates. We test that assumption on a complete three-factor design: ten transcription factors deleted in every single, double and triple combination in a free-fatty-acid-producing *Saccharomyces cerevisiae* strain, with all 720 deletion orders measured rather than inferred. Interaction is the rule. On total titer under a multiplicative null, 86 of 119 testable triples deviate at a 5% false discovery rate, and the median deviation moves from **+0.40** at two deletions to **−0.72** at three; each of the five measured species interacts too, and they disagree on the sign. Fifty-one triples exceed the base strain, yet only two sit at the end of a route whose every intermediate also improves, and the shallowest route falls a median 31% below the starting titer before recovering. None of the seven interaction networks tested is enriched among the interacting triples.

Keywords: metabolic engineering, epistasis, gene interactions, strain design, combinatorial deletion, *Saccharomyces cerevisiae*

Status: draft of a separate publication, not a working note. This is intended for submission as a standalone perspective once the free fatty acid study it re-analyzes is published, so it deliberately tracks the house style of the TorchCell manuscript ([paper/nature-biotech/editing.pdf](#)): same sn-jnl class, same sn-nature bibliography style, same print-approximate editing geometry, same section status chips. Keep it that way, and when a convention changes in [paper/nature-biotech](#), change it here too.

Target: *Metabolic Engineering*, as a research article. The finding is a statement about how build-and-screen campaigns are run, so the readership that acts on it is the one that designs those campaigns, and the journal already publishes analyses of combinatorial strain data. The re-analysis carries original work, the four nulls, the route enumeration and the model comparison, so it is not a commentary. Fallbacks, in order: *ACS Synthetic Biology*, then *Metabolic Engineering Communications*.

Two things are deliberately unfinished. Authorship is a placeholder and must include the authors of the source study. The bibliography is provisional: the document has no Zotero collection of its own, so `references.bib` is built from the served whole-library export, subset to the keys cited here, and only works already in the library can be cited.

✓ready ■author-reviewed, keep stable ✗not done, agents may edit freely

Contents

1	Introduction ✗	3
2	Results ✗	3
	A complete three-factor design over ten transcription factors	3
	Interaction is the rule, and the third deletion is where it bites	3
	Improving combinations are not reachable one deletion at a time	5
	What counts as an interaction depends on the null and on the scale	9
	No interaction network on hand predicts which combinations interact	9
	A model trained on growth interactions does not predict the titer interactions	11
3	Discussion ✗	12
	What theory predicts about where epistasis comes from	13
	What this design can and cannot support	13
	What would settle it	14
4	Methods ✗	14
	Data and normalization	14
	Epistasis models	14
	Uncertainty and multiple testing	15
	Path enumeration and monotonicity	16
	Interaction graph enrichment	16
	Scoring the triples with a growth-trained model	16
	Figures	16
	Code and data availability	16
	Supplementary Note 1: The digenic layer the trigenic score is measured against ✗	19
	Supplementary Note 2: The five species under the summed total ✗	20

1 Introduction x

A strain is improved by editing it, measuring it, keeping the edits that helped, and editing again. Every round of that loop is a decision made on the strain in hand, and a round that produced nothing better than its parent is a round whose output is discarded. The procedure is the design-build-test-learn cycle in its most common form, and its cost scales with the number of rounds rather than with the number of edits eventually carried.

The procedure is efficient exactly when the effects of edits compose. If a deletion that raises titer on its own keeps raising titer in the presence of other deletions, then a greedy walk reaches a good combination, and the number of strains that must be built grows linearly with the number of edits rather than exponentially. The assumption is rarely stated because it is rarely testable: a campaign that builds a few hundred strains out of a combinatorial space of millions cannot see the combinations it did not build, and cannot tell a combination that is genuinely bad from one it never reached.

Epistasis is the name for the failure of that assumption. In its quantitative form it is the deviation of a combination’s phenotype from what the phenotypes of its parts predict, and it is measured against a stated null [1]. Systematic genetic interaction mapping in *Saccharomyces cerevisiae* established both the scale of the phenomenon on growth rate and the estimator now used for it: a digenic interaction score defined as the departure of a double mutant’s fitness from the product of the two single-mutant fitnesses [2, 3], and a trigenic score that removes the digenic contributions before asking what the third deletion adds [4]. Those maps were built on fitness in rich medium, and the network they describe is itself environment dependent [5]. Whether their conclusions transfer to a production phenotype is not obvious, because fitness and titer are under different selection, are measured on different instruments, and need not share a null.

Combinatorially complete designs are what turn that question into a measurement, and they exist for other phenotypes: an enzyme active site [6], a set of chromatin regulators [7], and the fitness of natural variants [8, 9]. Metabolic engineering has the multiplexed editing tools to build such a design in a producing strain [10], and the case analyzed here is one.

What has been missing for a production phenotype is a design in which every combination and every construction order is present. Without one, a claim that a combination was unreachable is indistinguishable from a claim that nobody happened to build it. The free-fatty-acid dataset re-analyzed here supplies exactly that: ten transcription factors deleted in every single, double and triple combination in one producing background, which makes all 720 orders of deleting three genes available as measurements rather than as inferences. The design permits a question that a partial screen cannot ask. Not how many combinations interact, which any screen can estimate, but how many good combinations a stepwise campaign could have found.

2 Results x

A complete three-factor design over ten transcription factors

The dataset is a combinatorial deletion series in a free-fatty-acid-producing *S. cerevisiae* strain. Ten transcription factors (FKH1, GCN5, MED4, OPI1, RFX1, RGR1, RPD3, SPT3, TFC7 and YAP6) were deleted in every combination of one, two and three genes, giving 10 single, 45 double and 120 triple deletion strains alongside the undeleted base strain. The base strain is itself an engineered chassis, an FFA-overproducing *S. cerevisiae* carrying *pox1Δ faa1Δ faa4Δ*, and the ten transcription factor deletions are additional to those three. Every titer below is relative to that chassis, so a triple deletion strain carries six deletions in total. Six quantities were measured per strain: the titers of myristate (C14:0), palmitate (C16:0), stearate (C18:0), palmitoleate (C16:1) and oleate (C18:1), and their sum, which we refer to as total titer. All titers below are normalized to the base strain, so a value of 1 is no change and 2 is a doubling.

Completeness is what makes the design usable for the question at hand. Because every sub-combination of every triple was built, each triple can be reached by six distinct deletion orders, and the intermediate strain at each step of each order was measured rather than modeled. The design therefore contains 720 fully measured construction routes.

Only one of the ten single deletions raises titer at all. Deleting GCN5 gives 1.06×; the other nine lose ground, and deleting RPD3 costs nearly half the titer at 0.52× (Fig. 2a). Combinations behave very differently: 29 of the 45 doubles and 51 of the 120 triples exceed the base strain, and the best strain in the design is a triple at 2.05×. The gap between what single deletions do and what combinations do is the first sign that the effects are not composing.

Interaction is the rule, and the third deletion is where it bites

We scored each combination against a multiplicative null, in which the expected phenotype of a combination is the product of its parts’ phenotypes. For a double deletion the digenic score is the deviation from the product of the two singles,

$$\varepsilon_{ij} = f_{ij} - f_i f_j, \quad (1)$$

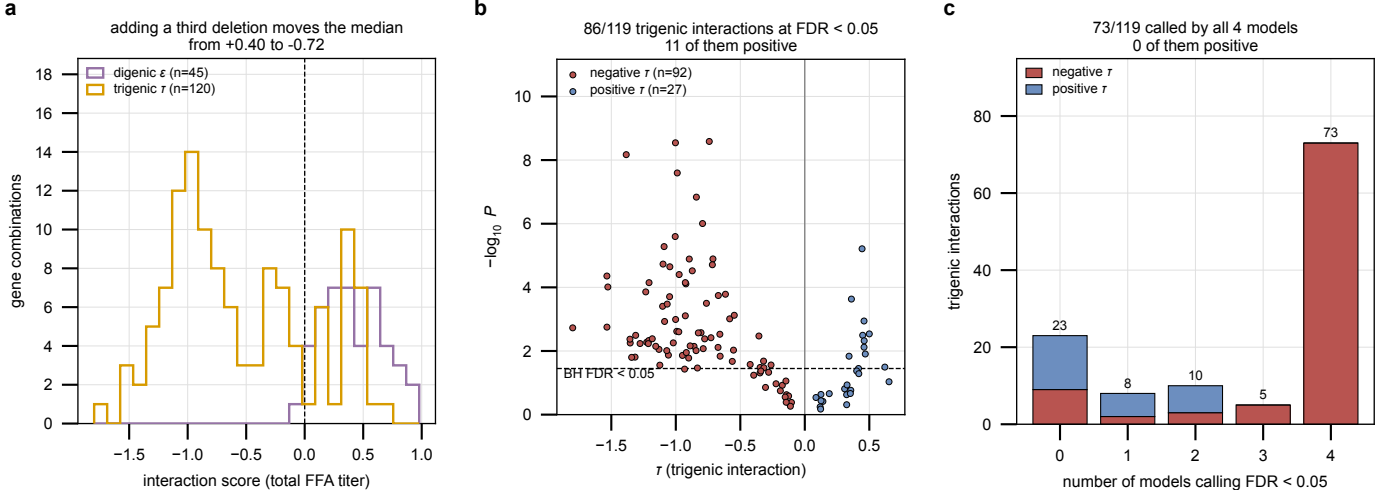


Fig. 1 Interaction is the rule on total fatty acid titer, and the third deletion is where it bites. **a**, Interaction scores against the multiplicative null for all 45 double (lilac) and 120 triple (amber) deletions, as outline histograms on shared bins so the heavy overlap stays readable. The median moves from +0.40 at two deletions to -0.72 at three. **b**, Trigenic interaction score τ against evidence for the 119 testable triples under the multiplicative model. The dashed line marks the largest P that Benjamini-Hochberg rejects at a 5% false discovery rate over the 119 tests of this readout ($P = 0.036$); 86 triples clear it and 11 of those have $\tau > 0$. **c**, How many of the four epistasis models call each triple significant, split by the sign of τ . Seventy-three triples are called by all four with the sign agreeing, and none of the 73 is positive.

and for a triple the trigenic score removes the digenic contributions before asking what remains:

$$\tau_{ijk} = f_{ijk} - f_{ij}f_k - f_{ik}f_j - f_{jk}f_i + 2f_i f_j f_k. \quad (2)$$

This is the trigenic score of the yeast genetic interaction maps [3, 4], applied to titer in place of fitness. Substituting Eq. 1 into Eq. 2 rewrites it as a residual against the pairs,

$$\tau_{ijk} = f_{ijk} - f_i f_j f_k - (\varepsilon_{ij} f_k + \varepsilon_{ik} f_j + \varepsilon_{jk} f_i), \quad (3)$$

an identity rather than an approximation. Read this way, τ is what the third deletion adds once the singles and the pairs have been given their due, and each pairwise deviation enters weighted by the remaining single-deletion phenotype. Standard errors propagate through Eq. 2 by the delta method, and each t statistic is referred to the Welch-Satterthwaite effective degrees of freedom of that linear combination, which has a median of 4.28 across the 714 testable trigenic scores.

Deviations from the null are large and are mostly negative at three deletions. The median digenic score is +0.40 and the median trigenic score is -0.72 (Fig. 1a), so adding a third deletion does not damp the interaction, it reverses its typical direction. Of the 119 triples with a testable standard error, 86 depart from the null at a 5% false discovery rate computed over the 119 tests of this readout, and 11 of those 86 depart upward (Fig. 1b).

The pairs are worth stating on their own, because they are what Eq. 3 measures the triples against. On total titer 43 of the 45 doubles have a positive digenic score, 25 clear the same false discovery rate and all 25 of those are positive, and 93 of the 120 triples produce less titer than their own singles and pairs predict (Supplementary Note 1). The reversal at the third deletion is therefore a change of direction rather than the continuation of a trend.

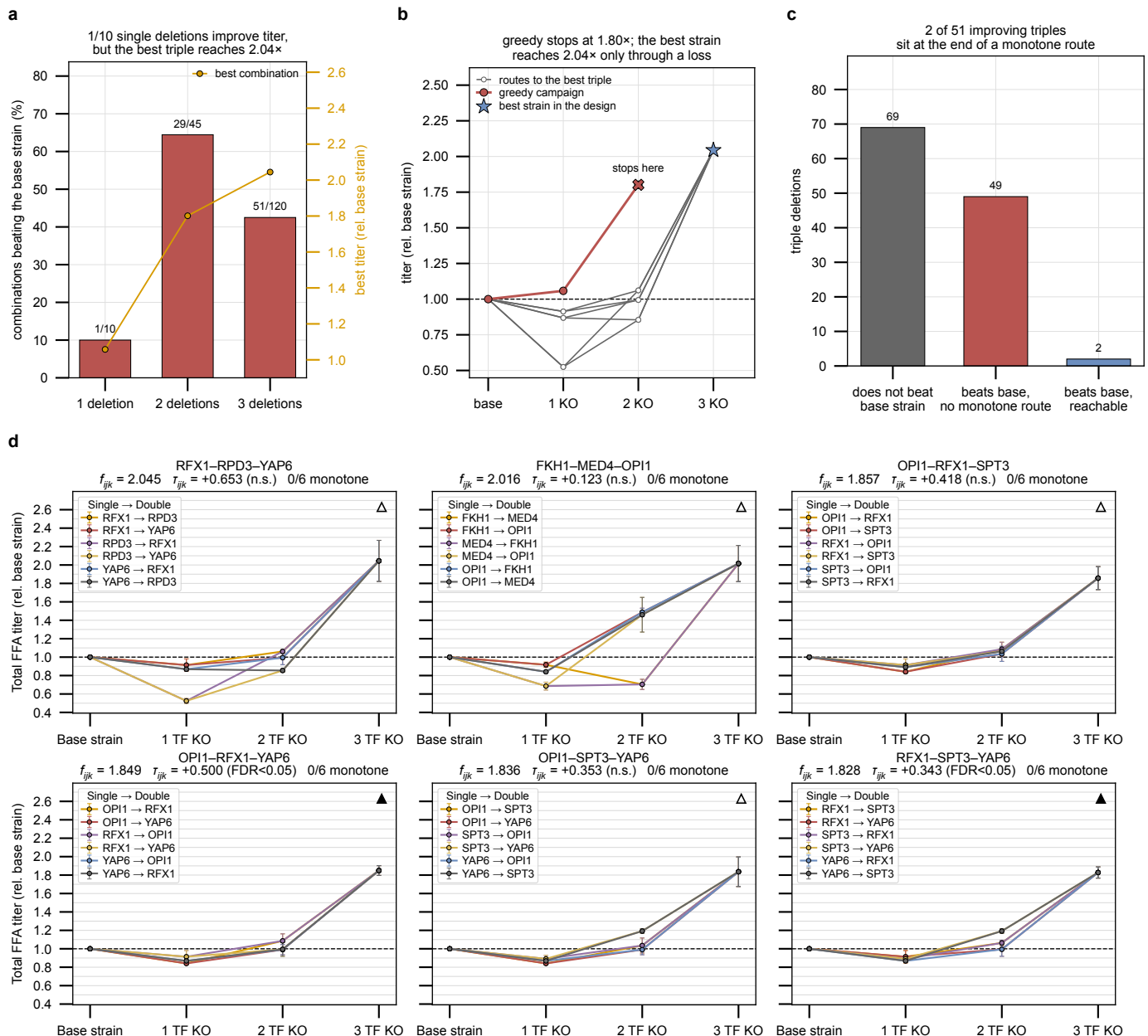


Fig. 2 Improving combinations exist, and almost none of them is reachable one deletion at a time. **a**, Percentage of combinations exceeding the base strain at one, two and three deletions (bars, counts above), with the best titer at each order (amber line, right axis). One of ten single deletions improves titer; the best triple reaches 2.05×. **b**, A greedy campaign (brick) takes the best strictly improving single deletion each round and stops when none improves. It reaches 1.80× after two deletions and halts. Gray lines are all six deletion orders of the highest-titer triple (RFX1, RPD3, YAP6; star), each of which passes through an intermediate below the base strain. **c**, The 120 triples classified by whether they beat the base strain and whether any of their six orders improves at every step. **d**, Trajectories for the six highest-titer triples, one colored line per deletion order, on a shared vertical scale. Rung 0 is the *pox1Δ faa1Δ faa4Δ* chassis and the three rungs after it add transcription factor deletions. Each panel gives that triple's final titer f_{ijk} and its trigenic score τ_{ijk} , and the badge marks the sign and significance of τ . None of the six has a monotone route. Error bars are standard errors of the strain means.

Improving combinations are not reachable one deletion at a time

An interaction score says a combination is surprising. It does not say whether a campaign could have found the combination, and that is the question a strain engineer is actually asking. We therefore scored each of the 720 measured routes for monotonicity: a route is monotone when every deletion in it leaves the strain strictly more productive than the strain before it, which is the condition under which a build-and-screen campaign that discards non-improving rounds would follow the route to its end.

Two of the 720 routes are monotone. Relaxing the criterion to allow a step whose loss is within one standard error raises the count to 11 routes across 9 triples, so the result is not an artifact of measurement noise sitting just the wrong side of zero. Counting by destination rather than by route: 51 of the 120 triples beat the base strain, and 2 of those 51 sit at the end of a monotone route (Fig. 2c). The other 49 improvements exist in the design and are invisible to a stepwise campaign.

The obstacle is a valley rather than a plateau. For each triple we took its shallowest route, meaning the one whose worst intermediate is highest, and measured how far that worst intermediate falls below the base strain. The median shallowest route dips 31% below the starting titer, and the deepest dips 48%. A campaign built to discard anything worse than its parent will discard the route at the first step.

Simulating the campaign directly gives the same answer in one number. Starting from the base strain and taking the best strictly improving single deletion each round, with full visibility of every candidate at every round, the campaign deletes GCN5 ($1.06\times$), then TFC7 ($1.80\times$), and then stops, because all eight remaining third deletions lower the titer (Fig. 2b). The best strain in the design is $2.05\times$, and reaching it requires passing through an intermediate at $0.91\times$ on its most forgiving order and $0.52\times$ on its least. The simulated campaign is more capable than a real one in every respect that matters, and it still recovers 88% of the achievable titer while stopping two rounds short of the answer.

The greedy trap has a second form worth naming, because it is not the same failure. One of the two monotone routes ends at FKH1, GCN5, MED4 at $1.73\times$. A campaign following it would improve at every round and finish with a triple. The simulated campaign does not find it, because at the second round TFC7 ($1.80\times$) beat FKH1 ($1.53\times$) and the better local step led to a dead end. Being reachable is necessary for a stepwise campaign to find a combination, and it is not sufficient.

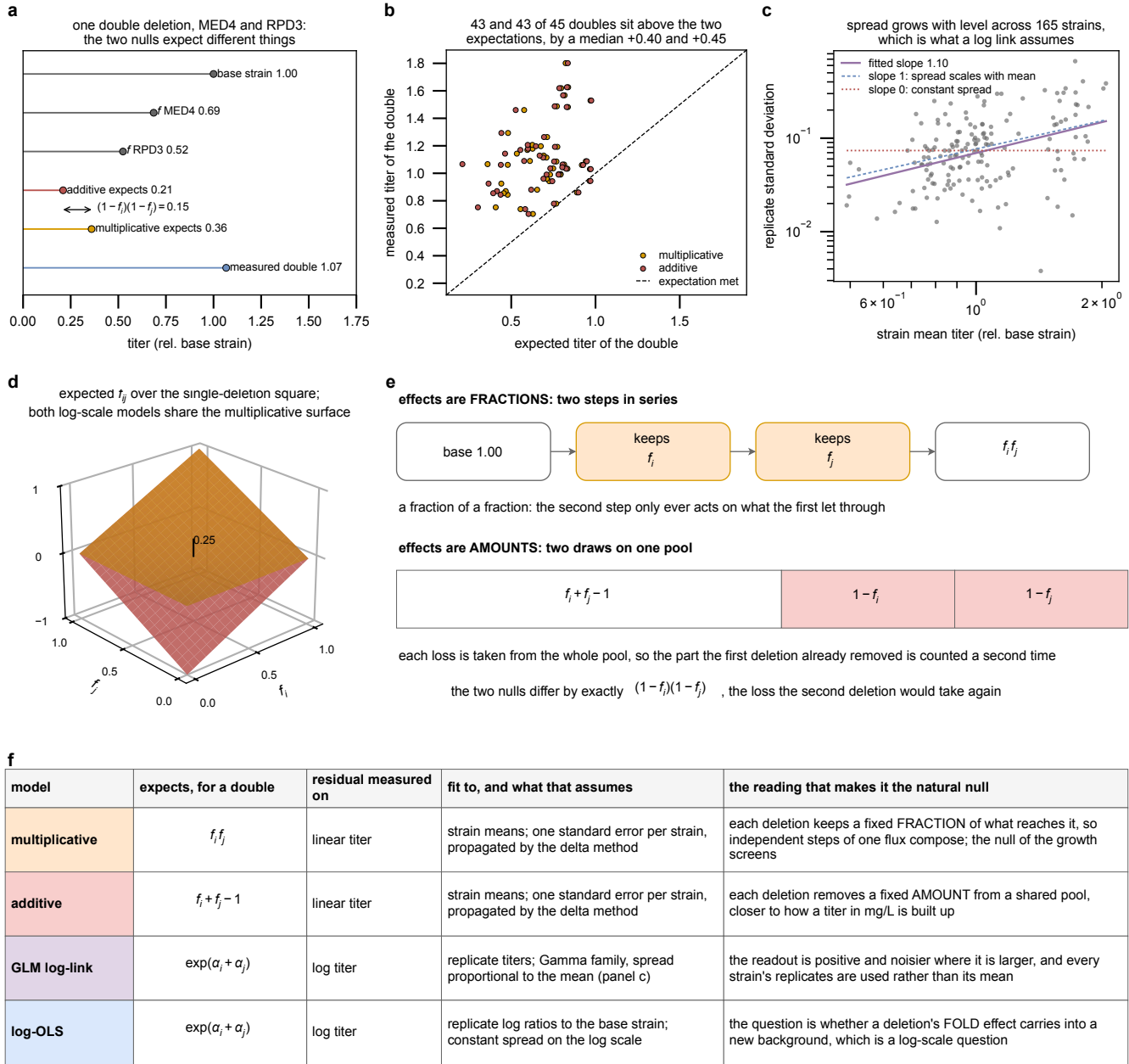


Fig. 3 What each null expects, and what choosing one commits to. **a**, One measured double deletion on a titer axis, MED4 and RPD3, the pair whose two expectations are furthest apart. The additive and the multiplicative expectation differ by $(1 - f_i)(1 - f_j)$, which is an identity rather than a property of this pair. **b**, All 45 doubles against both expectations; distance from the dashed line is the digenic interaction under that model. Forty-three of the 45 sit above both, by a median +0.40 against the multiplicative expectation and +0.45 against the additive one, so the two nulls agree on the direction at two deletions and differ in magnitude. **c**, Replicate standard deviation against strain mean over the 165 strains with a nonzero replicate spread, on log axes. The fitted slope is 1.10, against 0 for a constant spread and 1 for a spread proportional to the mean, which is the Gamma family and is the assumption behind a log link. **d**, The two expectations as surfaces over the square of single-deletion phenotypes. They meet along the two edges where one deletion does nothing and separate in the interior, by exactly $(1 - f_i)(1 - f_j)$, which is 0.25 at the center. Both log-scale models predict the same surface as the multiplicative one, so the four models differ there in the scale their residual is measured on rather than in what they expect. **e**, The two families of null as statements about the system. A multiplicative null says each deletion keeps a fraction of what reaches it, which is how independent steps of one flux compose and is the convention the growth interaction screens use. An additive null says each deletion removes an amount from a shared pool, and so counts the part of the first deletion's loss that the second would have taken again. **f**, The four models fit here: what each expects, the scale its residual is measured on, what it is fit to, and the reading of the system under which it is the natural null. Equations in the Methods.

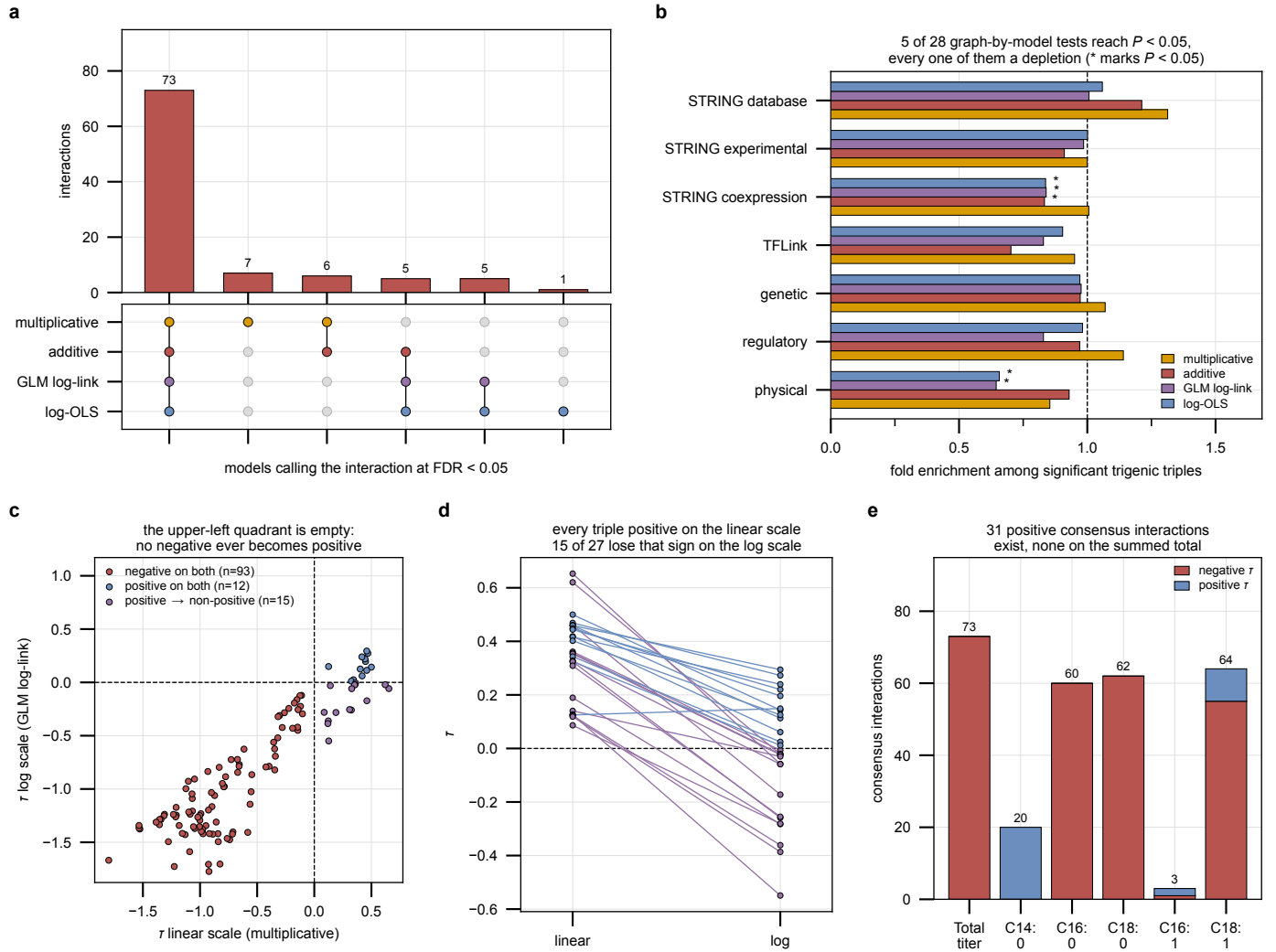


Fig. 4 What counts as an interaction depends on the null, on the scale and on the readout. **a**, Triples called significant at a 5% false discovery rate by each combination of the four models. Ninety-seven triples are called by at least one model and 73 by all four. **b**, Fold enrichment of seven interaction graphs among the significant trigenic triples, per model, over all six readouts pooled at $P < 0.05$. Asterisks mark $P < 0.05$ by Fisher's exact test; all five significant results are depletions. **c**, Trigenic score on the linear scale against the same quantity on the log scale. The upper-left quadrant is empty, so no combination that is negative on one scale is positive on the other. **d**, The 27 triples with a positive linear-scale score, carried onto the log scale; 15 lose the sign. **e**, Consensus interactions per readout, split by sign. Thirty-one positive consensus interactions exist across the five species, and none of them survives on their sum.

What counts as an interaction depends on the null and on the scale

Equation 2 is one choice among several, and a claim that rests on it should say how much of the claim is the choice. We fit three alternatives on the same normalized strain means: an additive null, in which expected phenotypes add rather than multiply; a generalized linear model with a log link and a Gamma family; and ordinary least squares on log titer differenced against the base strain. The four are not four estimates of one quantity. They encode different statements about how two deletions are expected to combine, and they measure their residuals on different scales (Fig. 3). A multiplicative null says a deletion keeps a fraction of what reaches it, which is how independent steps of a single flux compose and is why the growth interaction screens use it; an additive null says a deletion removes an amount from a shared pool, which is the reading a titer in milligrams per liter invites. The two differ by exactly $(1 - f_i)(1 - f_j)$ at two deletions, so they separate most where both single deletions are severe (Fig. 3a,d).

The four models agree on which combinations interact far more than they agree on the sign. Ninety-seven triples are called by at least one model at a 5% false discovery rate within the readout, 73 by all four, and every one of the 73 is negative (Fig. 4a). The disagreement falls almost entirely on the positive calls, and it falls along the scale: the multiplicative and additive models report 11 and 6 positive triples on total titer, and the two log-scale models report none.

That split is not a difference of power, it is a difference of estimand. The log-scale three-way interaction $g_{ijk} - g_{ij} - g_{ik} - g_{jk} + g_i + g_j + g_k$, with $g = \log f$, agrees with the linear-scale score in magnitude ($r = 0.91$, Spearman $\rho = 0.84$) but not near zero. Of the 27 triples with a positive linear-scale score, 15 are non-positive on the log scale, while no negative linear-scale score becomes positive (Fig. 4c,d). Positive three-way epistasis on fatty acid titer is a linear-scale statement, and reporting one without naming the scale is reporting the modeling choice as a result.

The readout matters as much as the scale. Thirty-one triples clear all four models with the sign agreeing on at least one of the five individual species, and 20 of those are on C14:0, where every consensus interaction is positive. None survives on the summed total (Fig. 4e). Summing the five species is a modeling choice too, and it is the choice that removes the positive signal. The five species behind the sum are not variations on one pattern: the number of triples called within a species runs from 3 to 80, C14:0 reverses sign between the digenic and the trigenic layer, and the summed total sits at $r = 0.91$ from C16:0 and C18:1 but at $r = -0.28$ from C14:0 (Supplementary Note 2). The four expectations themselves, and what each one costs to adopt, are written out in the Methods.

No interaction network on hand predicts which combinations interact

If interactions were predictable from prior knowledge, a campaign could avoid the problem by choosing combinations it expects to be additive. We tested that directly by asking whether the interacting triples are enriched for connectivity in existing interaction graphs. Seven graphs cover these ten genes with at least one connected triple: physical interactions, regulatory interactions, genetic interactions, TFLink, and the coexpression, experimental and database channels of STRING 12.0. For each graph and each model we compared the fraction of significant triples whose three genes form a connected subgraph to the same fraction among the non-significant triples.

Nothing is enriched. Across 28 graph-by-model tests, 5 reach $P < 0.05$ and every one of the 5 is a depletion rather than an excess (Fig. 4b). The largest excess anywhere is the STRING database channel under the multiplicative model at 1.31-fold, which does not reach significance ($P = 0.13$). Drawing the significant interactions onto the pathway itself makes the same point visually: the interacting triples span the transcription factor set with no visible relation to which factors regulate which fatty acid genes (Fig. 5).

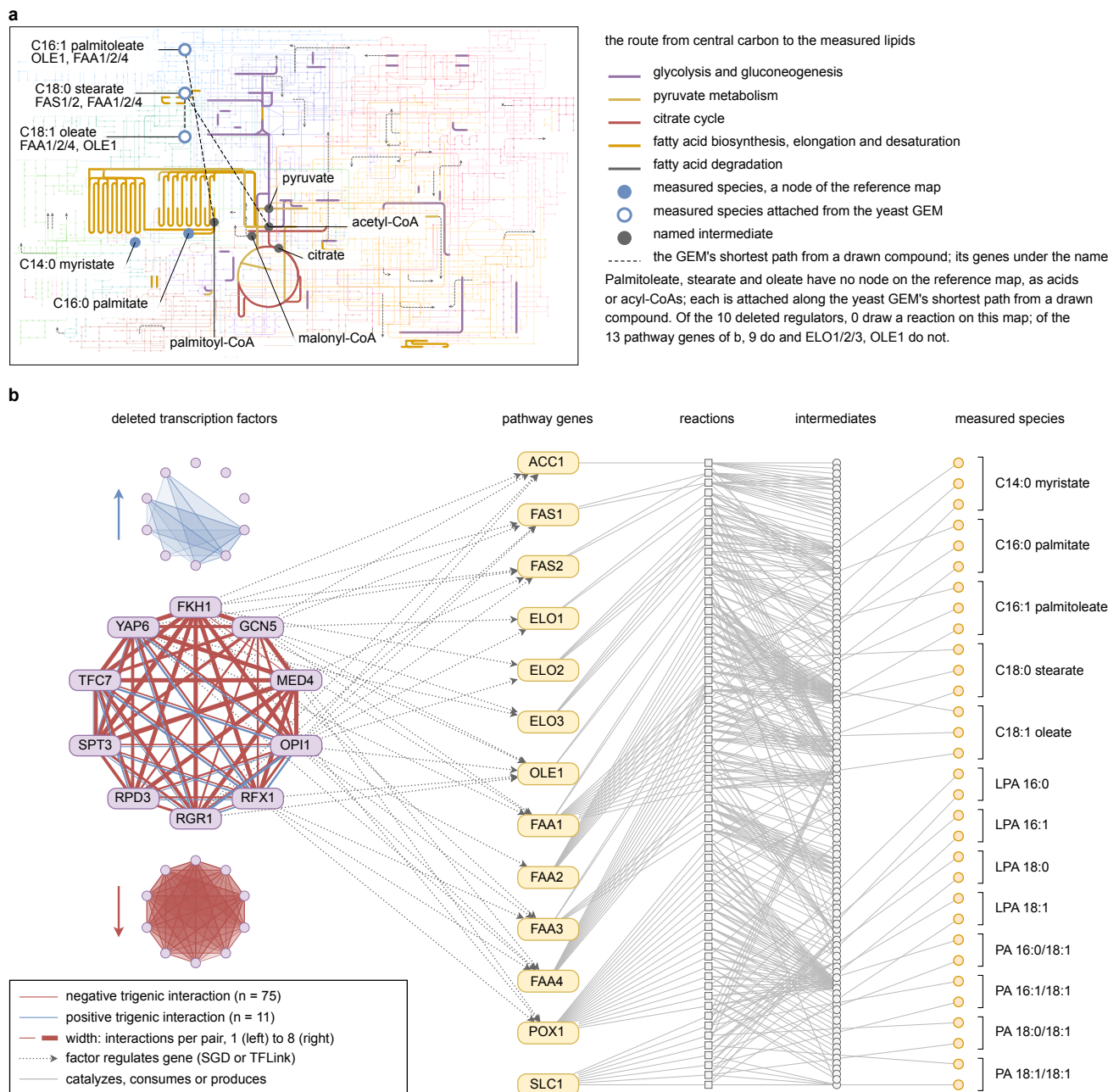


Fig. 5 Trigenic interactions on total fatty acid titer, drawn upstream of the pathway they act through. a, Where the measured chemistry sits in metabolism, on the whole of the iPath3 reference map of KEGG's global metabolic pathways. Colored are the yeast reactions of glycolysis and gluconeogenesis, pyruvate metabolism, the citrate cycle, fatty acid biosynthesis, elongation and desaturation, and fatty acid degradation, taken from KEGG's own gene lists for those pathways rather than chosen here. Filled blue nodes are the two measured species the reference map draws, myristate and palmitate; gray nodes are five named intermediates on the route. Palmitoleate, stearate and oleate have no node on that map, as free acids or as acyl-CoA thioesters, and the map likewise draws none of the four yeast elongation and desaturation genes (ELO1, ELO2, ELO3, OLE1) although KEGG's current global map lists them all: the gap is in iPath3's drawing, not in KEGG. So that both panels show the same five species, those three are attached as open rings along the shortest path of the yeast genome-scale model, the same model panel **b** is built from, each from the drawn compound that path starts at, with the genes of the path's reactions under its name. Of the ten deleted regulators, none has an orthology group drawn anywhere on this map, which is the point of the panel: the deletions act on metabolism from outside it. The map's own labels are not reproduced, because at this size they would print below the minimum type size. **b**, The ten deleted transcription factors are arranged in a circle at left. Edges within the circle are the 86 trigenic interactions significant at a 5% false discovery rate within the total-titer readout under the multiplicative model, 75 negative (brick) and 11 positive (blue), each drawn as the three edges of its triangle; edge width is the number of interactions the pair takes part in, from 0.40 pt at 1 to 2.15 pt at 8. Every one of the 45 pairs is in at least one negative interaction, so a pair that also carries a positive one would hide it: the two signs of a pair are therefore drawn as two parallel lines, offset perpendicular to the pair by half their combined width, and neither covers the other. Above and below that circle the same ten factors are drawn again at the same angles, unlabeled, carrying only the positive interactions (blue, above) and only the negative ones (brick, below). The two shapes are what the combined circle cannot show. The negatives reach all 45 pairs and every factor, with up to 8 interactions on one pair. The positives reach 17 pairs and 7 factors, at most 4 on a pair, and FKH1, GCN5 and MED4 take part in none of them. Dotted arrows are regulatory links from a factor to a pathway gene in the SGD regulatory graph or TFLink; three of the ten factors have none. The columns to the right are the 13 core fatty acid pathway genes, the 64 reactions they catalyze, the intermediates those reactions consume or produce, and the 31 measured species, one node per compartment (peroxisome, ER membrane, lipid particle) with a bracket per species. LPA, 1-acyl-sn-glycerol 3-phosphate; PA, phosphatidate, written as the two acyl chains.

A model trained on growth interactions does not predict the titer interactions

The other place a campaign could look for a predictor is a model trained on measured combinations rather than on a graph. The torchcell equivariant cell graph transformer, described in a companion manuscript in preparation ([paper/nature-biotech/editing.pdf](#)), is trained on the trigenic growth interaction screens of Kuzmin et al. [4, 11], where each triple deletion is scored by colony fitness in a wild-type background. What that model is measured against, and how it behaves on the additive nulls, is worked through in [12]. All ten factors here are in its gene space, and seven of them were deleted somewhere in its training data; GCN5, RPD3 and TFC7 were not. We scored each of the 120 triples as a three-gene deletion and asked whether the predicted growth interaction tracks the measured titer interaction under the multiplicative model (Fig. 6). The two quantities differ in readout and in background, so a null here does not separate a model that cannot rank trigenic interactions from a growth interaction that does not carry over to titer. The question is only whether this predictor, as it stands, would help choose combinations.

Neither magnitude nor rank transfers. Across the 120 triples the ensemble mean of three checkpoints correlates with the measured τ at $r = 0.16$ ($P = 0.08$) and Spearman $\rho = 0.13$ ($P = 0.17$), and the measured τ shows no trend across quartiles of the prediction (Kruskal-Wallis $P = 0.39$). The three checkpoints agree with each other (r 0.62 to 0.71) far better than any of them agrees with titer (r 0.08 to 0.20), so the miss is a property of what the model learned rather than of one training run. Restricting to the 86 triples called at a 5% false discovery rate, to the 84 free of the contested TFC7 locus, or to the 35 whose three factors were all deleted in training moves r between 0.12 and 0.16 without moving it out of the noise, and the top quartile by predicted τ holds no more called interactions than the rest (22 of 30 against 64 of 90).

The one signal is the sign. Of the 20 triples the model predicts positive, 9 measure positive, against 18 of the 100 it predicts non-positive: 45% against 18% (Fisher $P = 0.016$). The same split holds in the 84 triples without TFC7 ($P = 0.005$) and in the 35 fully trained ones ($P = 0.04$). Every P in this subsection is nominal among ten related tests on one set of 120 triples, so the sign association is a lead to test on a new panel, not a result. Read with the graph result, the training data a useful predictor needs are combinations measured on the readout of interest: a model trained on growth combinations carries over to titer at most the direction of the interaction.

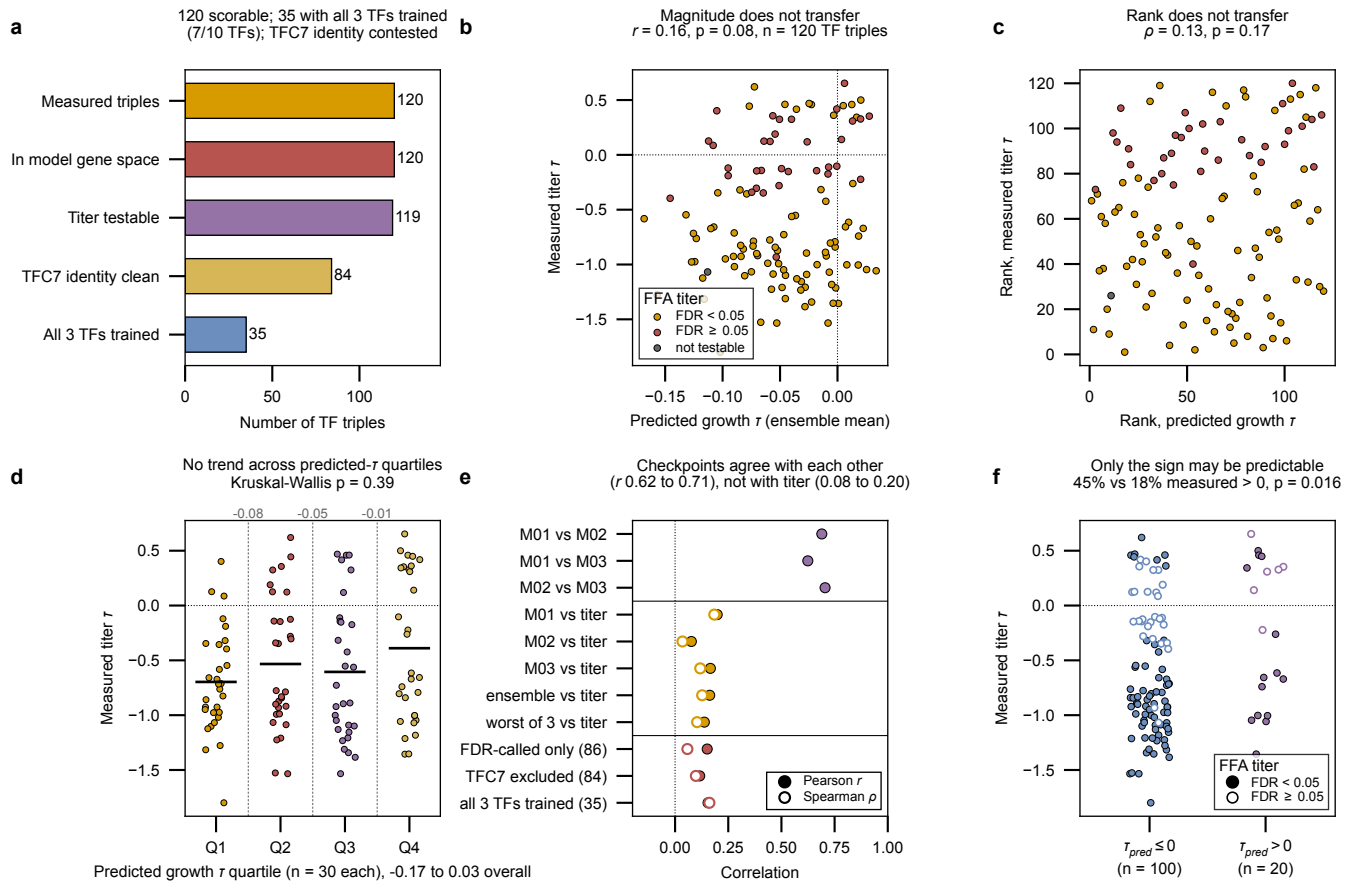


Fig. 6 A growth-trained transformer does not predict the titer interactions. Three checkpoints of the torchcell equivariant cell graph transformer, trained on the Kuzmin 2018 and 2020 trigenic growth screens, scored each transcription factor triple as a three-gene deletion in a wild-type background; the measured quantity is the multiplicative titer interaction τ of the same triple on the *pox1Δ faa1Δ faa4Δ* chassis, total titer. “FFA titer FDR” is Benjamini-Hochberg over the 119 testable triples within total titer. **a**, Coverage: 120 scorable, 35 with all three factors seen as deletions in training (7 of 10 factors), 84 free of the contested TFC7 locus. **b**, Ensemble-mean predicted growth τ against measured titer τ , Pearson $r = 0.16$, $P = 0.08$; the key’s classes number 86, 33 and 1. **c**, The same as ranks, Spearman $\rho = 0.13$, $P = 0.17$. **d**, Measured τ by quartile of predicted τ , bar at the bin mean, Kruskal-Wallis $P = 0.39$. The dashed dividers are the three cut points between quartiles, labeled with the predicted τ at each. **e**, Pearson (filled) and Spearman (open) correlations: between checkpoints on the same triples; each checkpoint, the ensemble mean and the worst of three against titer; the ensemble on three subsets. **f**, Measured τ split by predicted sign (blue at or below zero, purple above; filled where the titer interaction is FDR-called), 45% against 18% measured positive, Fisher $P = 0.016$. Every P is nominal among ten related tests on one set of 120 triples.

3 Discussion x

The design space of a strain engineering campaign is usually described as too large to search. Ten deletable targets already give 1,023 combinations, and a genome-scale target list gives more combinations than there are strains anyone will ever build. That framing puts the difficulty in the size of the space and implies that the remedy is throughput or a better search.

The measurements here point at a different difficulty, which throughput does not fix. In a space small enough to enumerate completely, with every combination and every construction order measured, the best strain is still unreachable by the procedure that campaigns actually use. The obstacle is not that the good combination is hard to find among many. It is that the good combination sits behind an intermediate that a screening rule would throw away. Of 51 improvements, 2 are reachable. Adding throughput multiplies the number of routes a campaign can try; it does not make a route monotone.

Three consequences follow for how a campaign is designed.

Retain losses. A rule that keeps only strains better than their parent discards the approach to almost every improvement in this design. Carrying a fixed number of non-improving intermediates forward per round costs build capacity in proportion, and it is the only change to the procedure itself that addresses the reachability problem. How many to carry is set by the depth of the valleys, which here have a median of 31% of the base titer.

Build combinations, not sequences. If the intermediates are uninformative about the endpoint, then building them is a cost with no return, and a campaign is better served by constructing the combinations it wants directly. That

is a statement about what multiplexed editing buys: not speed at reaching a known target, but access to targets that no sequence of single edits reaches.

Predict the combination, not the parts. A model that predicts a combination’s phenotype from its parts’ phenotypes fails on exactly the combinations that matter here, because those are the combinations where the parts do not compose. A useful predictor has to be trained on combinations, and the enrichment result says the existing interaction graphs will not stand in for that training data: across 28 graph-by-model tests, the interacting triples are never enriched for connectivity in any graph, and are significantly depleted in five. Whatever determines which combinations interact is not recorded in physical, regulatory, genetic, TFLink or STRING connectivity among these ten genes. Nor is it carried by combinations measured on a different readout: a transformer trained on trigenic growth interactions predicts neither the magnitude nor the rank of the titer interactions, and at most their sign. The combinations a predictor learns from have to be measured on the phenotype it is asked to predict.

What theory predicts about where epistasis comes from

There is a body of theory on when deletions in a metabolic system should compose and when they should not, and it is worth asking what it predicts here. Flux balance analysis over the yeast metabolic network gives a trimodal distribution of pairwise epistasis and organizes the interactions into modules that interact monochromatically, meaning that the links between two functional modules are either all buffering or all aggravating [13]. A complementary result derives how epistasis propagates in a hierarchical metabolic network with first-order kinetics, and finds that weak epistasis at a lower level is distorted as it moves to higher-level phenotypes, so pairwise interactions between genes should be common and should depend on the genetic background [14]. Both predict what is observed here at the level of prevalence. Interaction is common, and its sign is consistent within a readout.

What neither predicts is the reachability result, and the reason is that both are theories about enzymes in a flux network rather than about transcription factors upstream of one. A factor deletion changes the expression of many pathway genes at once, and it also changes genes outside the pathway, so the module structure a flux model assigns to the pathway is not the structure the deletions act on. The one systematic study of epistasis between gene-specific transcription factors in yeast reached its mechanisms through expression profiling of the double deletions rather than through pathway topology [15], and proposed two routes, buffering by induced dependency and alleviation by derepression, that connect genes in parallel pathways with no shared reaction. The enrichment result here is consistent with that: connectivity in a physical, regulatory, genetic or coexpression graph does not predict which of these triples interact. Applying the metabolic theory to a factor panel would require a model that maps a factor deletion to a distribution of flux changes first, and that model is what the field does not yet have for titer.

What this design can and cannot support

The design is complete, which is what licenses the reachability claim, and it is small, which bounds it. Ten transcription factors in one producing background, one medium and one harvest time do not establish a rate at which improvements are unreachable in general. They establish that the rate can be 2 in 51 in a real production campaign’s target set, which is enough to make the assumption of composability something that has to be checked rather than assumed.

Two features of the system may not transfer. The deleted genes are transcription factors rather than pathway enzymes, so a deletion acts on many pathway genes at once and the interactions may be denser than a comparable enzyme panel would give. And the phenotype is a titer under production conditions rather than growth rate, so the deletions are under no selection to be individually tolerable. Both cut toward more interaction here than a reader should expect elsewhere. A single-gene-per-step campaign on enzymes may well find a monotone route more often. What the design shows is that the question has an answer, and that the answer is measurable rather than assumable.

The measurement itself carries one limitation that applies to every number in this document. Each strain contributes an endpoint titer in milligrams per liter from three replicate cultures, and the released data include no biomass measurement, so nothing here is normalized to cell density. A deletion that lowers final biomass and a deletion that lowers fatty acid output per cell are indistinguishable in this readout, and both appear as a lower f . Two consequences follow and neither is removable by reanalysis. Interaction scores mix specific productivity with growth, so a triple whose three deletions each cost biomass could show a negative τ with no change in per-cell output. And the valley depths that drive the reachability result are titer valleys, which is the right quantity for a campaign that screens on titer, and is not evidence about the yield or the rate. What the replicates do support is that the values are stable: the trigenic standard errors propagate from three replicates per strain through seven strains, and the monotone-route count moves only from 2 to 11 when a step is allowed to lose up to one standard error, so the result is not sitting on measurement noise. A design that reported optical density alongside titer would separate the two routes, and that is a property of the source experiment rather than of the analysis.

What would settle it

Answering the general question needs more datasets of the same shape, and the shape is restrictive. A partial combinatorial screen cannot support a reachability claim at all, because an unbuilt intermediate is indistinguishable from a bad one. The requirement is a design in which every sub-combination of every reported combination was itself built and measured, in the same background and the same batch. Screens that report only the top performers, or only the combinations that a campaign happened to reach, do not qualify no matter how many strains they contain.

The complementary route is theory. A production phenotype is the output of a metabolic network under constraints, and a network model that predicts where in a pathway deletions compose and where they do not would turn the reachability question into a calculation performed before the strains are built. The pieces exist for enzymes in a flux network [13, 14], and what is missing is the step from a regulator deletion to the flux changes it causes. That model does not exist yet for titer, and the enrichment result above says it will not be recovered from connectivity alone. Building it is the direction this line of work points.

4 Methods ✗

Data and normalization

Titers for five fatty acid species and their sum were taken from the source study’s supplementary tables for the base strain and for all 175 deletion combinations. Each strain contributes three replicate measurements per species, in milligrams per liter at a single endpoint. Every titer is divided by the base strain’s titer for the same species in the same experiment, so all values below are ratios and the base strain is 1 by construction. The sum, referred to as total titer, is formed per replicate before averaging rather than by summing the five published species means, so it is a measured quantity and not a sixth derived one. Loading and normalization are in `experiments/008-xue-ffa/scripts/free_fatty_acid_interactions.py`.

The released measurements carry no biomass or optical density column, so no quantity in this document is normalized to cell number. Every f is a volumetric titer ratio, and a deletion that lowers final biomass is not separable from one that lowers output per cell. This is a property of the released data rather than a choice made here, and it bounds what the interaction scores mean: they are interactions on volumetric titer, which is the quantity a titer-screened campaign selects on.

Epistasis models

An interaction score is a residual, and a residual is only defined once the expectation is written down. The four models below write down four different expectations for the same seven measured strains, so a triple can be an interaction under one and not under another without either fit being wrong. Figure 3 draws the four side by side on one worked pair of deletions.

Setting. Write f_S for the titer of the strain carrying deletion set S , divided by the titer of the base strain in the same experiment, so $f_\emptyset = 1$ by construction. Single, double and triple deletions are f_i , f_{ij} and f_{ijk} . All four models are fit on these same normalized quantities, and none of them sees a combination the design did not build.

Multiplicative. The expected phenotype of a combination is the product of its parts, which is the null of the yeast genetic interaction maps:

$$\varepsilon_{ij}^\times = f_{ij} - f_i f_j, \quad \tau_{ijk}^\times = f_{ijk} - f_{ij} f_k - f_{ik} f_j - f_{jk} f_i + 2 f_i f_j f_k. \quad (4)$$

Independence on this scale means that a deletion multiplies titer by the same factor whatever else is deleted. These are Eqs. 1 and 2, and this is the model every figure uses unless it says otherwise (`free_fatty_acid_interactions.py`).

Additive. The same construction with sums in place of products:

$$\varepsilon_{ij}^+ = f_{ij} - f_i - f_j + 1, \quad \tau_{ijk}^+ = f_{ijk} - (f_{ij} + f_{ik} + f_{jk}) + (f_i + f_j + f_k) - 1. \quad (5)$$

Independence here means that a deletion shifts titer by the same amount whatever else is deleted. The +1 and -1 are the base strain entering at $f_\emptyset = 1$ (`additive_free_fatty_acid_interactions.py`).

GLM log-link. A generalized linear model with a Gamma family and a log link, fit on the replicate-level titers rather than on strain means, with $x_i \in \{0, 1\}$ marking deletion i and r indexing replicate:

$$\log \mathbb{E}[y_{S,r}] = \mu + \sum_i \alpha_i x_i + \sum_{i < j} \beta_{ij} x_i x_j + \sum_{i < j < k} \gamma_{ijk} x_i x_j x_k + \rho_r. \quad (6)$$

The interaction is γ_{ijk} . The base strain is excluded from the fit and enters through μ . Gamma sets the variance proportional to the square of the mean, which is the right shape for a titer that cannot go negative and whose spread grows with its level (`glm_log_link_epistatic_interactions.py`).

Log-OLS. Ordinary least squares on log titer differenced against the base strain within each replicate, with $\delta = 10^{-6}$ a pseudocount:

$$g_{S,r} = \log(y_{S,r} + \delta) - \log(\bar{y}_{\emptyset,r} + \delta), \quad (7)$$

regressed on the same saturated design as Eq. 6. Its three-way coefficient estimates the saturated log-scale interaction

$$\tau_{ijk}^{\log} = g_{ijk} - g_{ij} - g_{ik} - g_{jk} + g_i + g_j + g_k, \quad g_S = \log f_S, \quad (8)$$

and that equivalence, which Fig. 4c,d rests on, is checked in code rather than assumed: `linear_vs_log_scale_sign_test.py` recomputes Eq. 8 from the strain means and raises unless it agrees with both fitted log-scale models at $r > 0.99$ (`log_ols_wt_differencing_epistatic_interactions.py`).

The scale is the first tradeoff, and it is not a matter of taste. Equations 4 and 8 are different quantities, not two estimates of one quantity. They agree in magnitude across the 119 testable triples at $r = 0.91$ and Spearman $\rho = 0.84$ and disagree near zero, which is exactly where a call is made: 15 of the 27 triples with a positive linear-scale score are non-positive on the log scale, and none of the reverse. A three-way interaction is not invariant under a monotone transform of the phenotype, so a positive trigenic interaction on titer is a linear-scale statement and has to be reported as one. Which scale is appropriate depends on what the reader will do with the number. A campaign that asks how many milligrams per liter a third deletion adds is asking a linear-scale question; a campaign that asks whether a deletion’s fold effect is preserved in a new background is asking a log-scale one.

The second tradeoff is what carries the pairwise information. The multiplicative trigenic score can be written with the digenic scores in it, which is Eq. 3 and is an identity rather than an approximation. Each pairwise deviation enters weighted by the remaining single-deletion phenotype, so a pair that interacts strongly in a background where the third deletion is nearly neutral contributes almost its full ε , and the same pair contributes little when the third deletion is severe. The log-scale interaction in Eq. 8 weights every term equally. Neither weighting is more correct; they encode different ideas of how a pair is expected to act in the presence of a third deletion, and the weighted form is the one that keeps comparability with the published yeast interaction maps.

The third tradeoff is power against assumption. The two linear-scale models are fit on strain means and propagate the replicate standard errors through Eqs. 4 and 5 by the delta method, so they assume nothing about the shape of the noise beyond a standard error per strain, and they pay for it in degrees of freedom: a trigenic score combines seven independently measured strains at three replicates each, giving a median Welch-Satterthwaite effective degrees of freedom of 4.28. The two regression models are fit on the replicate-level values and borrow strength across the whole design, which raises power and buys a mean-variance assumption in exchange, Gamma for Eq. 6 and homoscedastic on the log scale for Eq. 7. A count of significant interactions is therefore not comparable across the two families without saying which is which, and that is why the Results report agreement on the identity of the interacting triples separately from agreement on their sign.

What this document uses, and why. The multiplicative model is the reported one throughout, for comparability with the trigenic screens the field already has, and every claim that rests on a sign is stated with the scale attached. The other three exist here to answer how much of a claim is the modeling choice, which is a question the reader can only settle if all four expectations are on the page.

Uncertainty and multiple testing

Standard errors propagate through the interaction expression by the delta method, using the per-strain standard errors of the replicate means. The resulting t statistic is referred to the Welch-Satterthwaite effective degrees of freedom of the linear combination, computed from each term’s contribution to the combined standard error. This matters more than it usually does. A trigenic score combines seven independently measured strains, and referring its t statistic to the degrees of freedom of the smallest single input, as an earlier version of this analysis did, gives 1 or 2 rather than a median of 4.28, and suppresses nearly every call. Across the 990 multiplicative tests of the whole run, none was significant at a 5%

false discovery rate before the correction and 456 are after it (audit record: `notes/plan.008-xue-ffa-epistasis-audit.2026.08.15.md`).

Benjamini-Hochberg controls the false discovery rate. The family matters and is stated wherever a count appears. Counts in Figs. 1 and 4 correct over the tests of a single readout, since that is what those figures claim about. The enrichment analysis in Fig. 4b uses the stored per-test $P < 0.05$ over all six readouts pooled, which is the family the enrichment scripts were written against.

Path enumeration and monotonicity

For each triple, all six orders of introducing its three deletions were expanded into a four-rung trajectory: base strain, one deletion, two deletions, three deletions, taking the measured strain at each rung. A route is monotone when every step is strictly positive, and monotone within one standard error when every step exceeds the negative of that step's propagated standard error. Valley depth is $1 - \min(\text{rungs})$ on a route, and a triple's valley depth is the minimum over its six routes, meaning its shallowest available approach. The greedy campaign in Fig. 2b starts at the base strain, evaluates every combination reachable by one further deletion, takes the highest if it exceeds the current strain, and stops otherwise. Enumeration and monotonicity are in `ffa_epistatic_path_panels.py`; the greedy simulation is in `perspective_figure_panels.py`.

Interaction graph enrichment

Ten interaction graphs were queried for each triple: physical, regulatory and genetic interactions, TFLink, and the neighborhood, fusion, co-occurrence, coexpression, experimental and database channels of STRING 12.0. A triple counts as overlapping a graph when its three genes induce a connected subgraph of that graph. Fold enrichment is the ratio of the overlapping fraction among significant triples to the overlapping fraction among the rest, and significance is Fisher's exact test on the corresponding two-by-two table. Three of the ten graphs contain no connected triple among these ten genes at all and are omitted from Fig. 4b, which is a statement about those graphs' coverage of this gene set and not about epistasis (`interaction_graph_enrichment_analysis.py`).

Scoring the triples with a growth-trained model

Each of the 120 triples was scored by the torchcell equivariant cell graph transformer trained on the Kuzmin 2018 and 2020 trigenic growth screens, as a three-gene deletion in a wild-type background, the perturbation form the model was trained on. The three chassis deletions were not part of the scored genotype, and no single or double was scored. Three independently trained checkpoints were scored on CPU with a parity guard against stored GPU predictions of the same genotypes (18 values, largest deviation 0.0012 at a tolerance of 0.005), and the prediction is the ensemble mean of the three. The predicted quantity is the trigenic growth interaction τ on the synthetic genetic array selection medium. It was compared with the measured multiplicative titer interaction by Pearson and Spearman correlation, by a Kruskal-Wallis test of measured τ across quartiles of the prediction, by Fisher's exact test of the predicted sign against the measured sign, and by Fisher's exact test of the top predicted quartile against the false-discovery-rate call (`experiments/010-kuzmin-tmi/scripts/ffa_panel_model_score.py`, tables in `experiments/010-kuzmin-tmi/results/ffa_panel_model_score.{csv,json}`).

Figures

Every panel is produced by a committed script and written as a true-size SVG at one of the standard panel widths, then converted to PDF for this document. Panels come from `perspective_figure_panels.py`, `notable_interaction_selection.py` and `ffa_epistatic_path_panels.py`; the supplementary panels and the per-readout count table behind Supplementary Note 1 and Supplementary Note 2 come from `supplementary_figure_panels.py`; the network overlay comes from `ffa_network_overlay_panel.py`, and `ffa_network_overlay_drawio.py` writes the same layout as native draw.io shapes, which is the source Fig. 5 is exported from; the growth-model figure comes from `experiments/010-kuzmin-tmi/scripts/ffa_panel_model_score.py`. The multi-panel arrangements are assembled from the panel SVGs by `build_perspective_drawio.py` into editable draw.io sources under `notes/assets/drawio/`.

Code and data availability

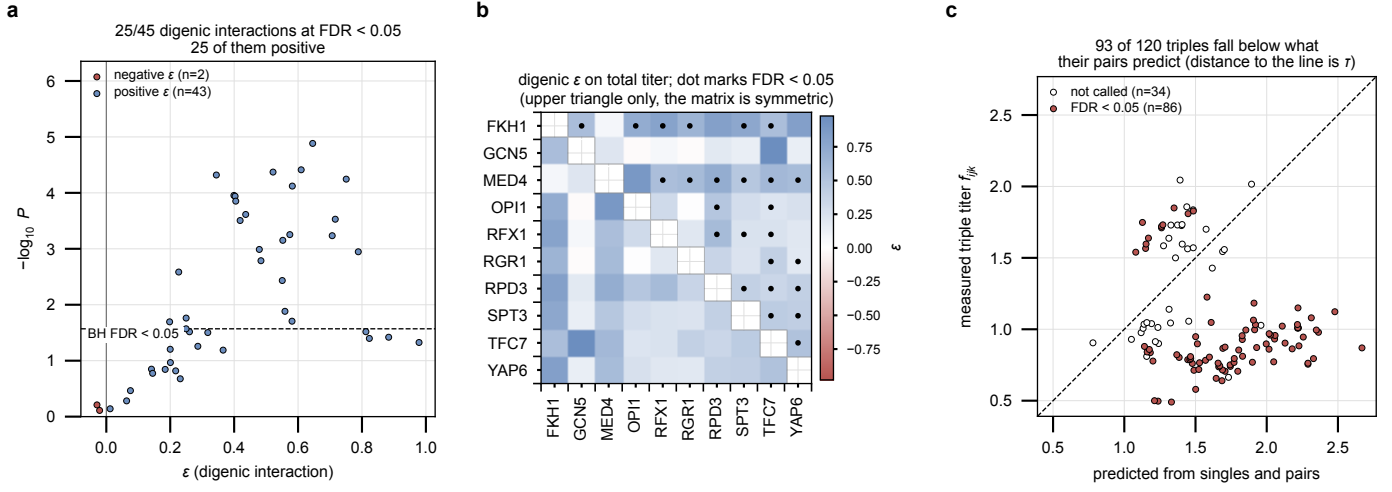
All analysis scripts are in `experiments/008-xue-ffa/scripts/` and all derived tables in `experiments/008-xue-ffa/results/` of the torchcell repository. Selection of which interactions appear in a figure is itself a committed artifact

rather than a post-hoc choice: `notable_interaction_selection.py` states the rule, applies it, and writes the selected set to `results/notable_interaction_selection.csv`.

Supplementary Information x

Contents

- Supplementary Note [1](#) The digenic layer the trigenic score is measured against (with Supplementary Fig. [1](#))
- Supplementary Note [2](#) The five species under the summed total (with Supplementary Fig. [2](#))
- Supplementary Figs. A figure supporting a Note sits with that Note



Supplementary Figure 1 The digenic layer on total fatty acid titer. **a**, Digenic score ϵ against evidence for the 45 double deletions under the multiplicative model. The dashed line marks the largest P that Benjamini-Hochberg rejects at a 5% false discovery rate over the 45 tests of this readout; 25 pairs clear it and all 25 are positive. **b**, Every pair's ϵ as a matrix over the ten factors, blue positive and brick negative on a scale centered at zero. The matrix is symmetric and the dots marking a false-discovery-rate call are drawn in the upper triangle only. **c**, Each triple's measured titer against the titer predicted from its singles and its pairs, $f_i f_j f_k + \epsilon_{ij} f_k + \epsilon_{ik} f_j + \epsilon_{jk} f_i$. The vertical distance to the diagonal is the trigenic score τ , so a point below the line is a triple that produces less than its own pairs predict; 93 of 120 do. Filled points are triples called at a 5% false discovery rate within total titer.

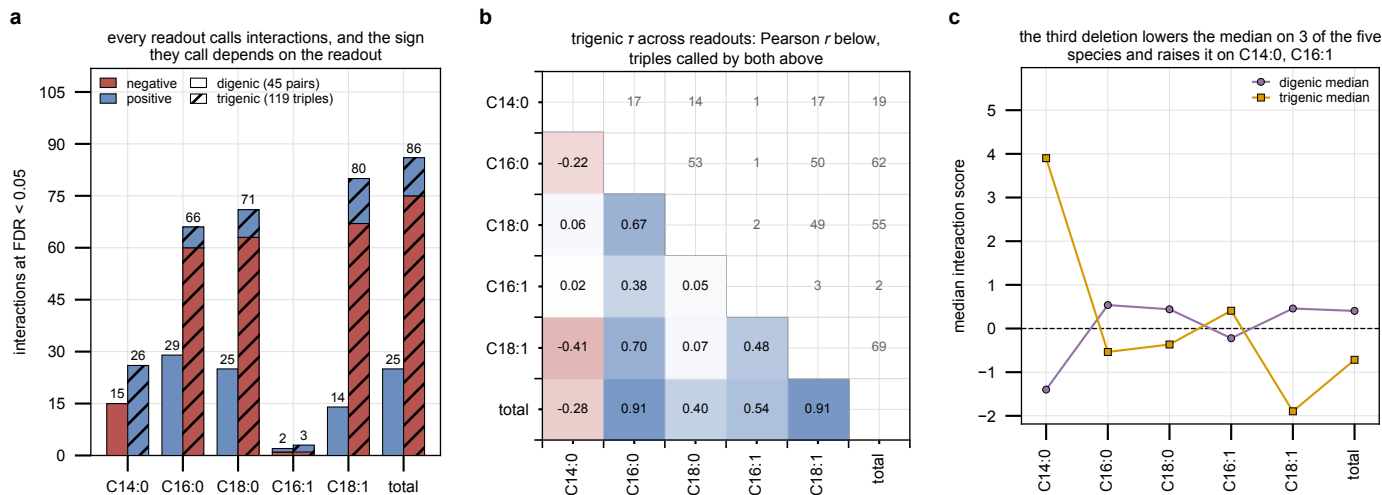
Supplementary Note 1: The digenic layer the trigenic score is measured against $\times$

The main text is a claim about the third deletion. That claim is made against what the first two do, so what the pairs do is worth stating on its own, and it is the opposite of the trigenic result.

The pairs are positive. On total titer, 43 of the 45 double deletions have a positive multiplicative digenic score and the median is +0.40. Twenty-five of the 45 reach a 5% false discovery rate within the readout, and every one of the 25 is positive (Supplementary Fig. 1a). A pair of deletions therefore tends to produce more titer than the product of its singles, which is the behavior a campaign would want and would find if it stopped at two.

No factor carries it. Drawing all 45 scores as a matrix over the ten factors (Supplementary Fig. 1b) shows the positive scores spread across the panel rather than concentrated in the rows of one or two factors. The largest score in absolute value is 0.98. Every one of the ten factors takes part in at least one called pair, from GCN5 in 1 of its 9 to TFC7 in 8 of its 9, so no single factor carries the positive digenic signal and none is exempt from it.

The third deletion undoes it. Writing the multiplicative expectation for a triple with the pairs included, as in Eq. 3, gives a prediction of the triple from its singles and its pairs alone. Ninety-three of the 120 triples fall below that prediction (Supplementary Fig. 1c), and the vertical distance to the diagonal in that panel is τ . The negative trigenic median reported in the main text is therefore not a continuation of a downward trend the pairs already set. The pairs trend upward, and the third deletion reverses the direction.



Supplementary Figure 2 The five measured species behind the summed total. **a**, Interactions called at a 5% false discovery rate within each readout, digenic (solid) and trigenic (hatched), stacked by sign, blue positive and brick negative. Counts above each bar are the totals. **b**, Trigenic τ compared across readouts over the same 119 triples: Pearson r below the diagonal, and above it the number of triples called at a 5% false discovery rate in both readouts. **c**, Median interaction score per readout for both orders. The gap between the two lines is the quantity the main text reports on total titer, and its sign is not the same on every species.

Supplementary Note 2: The five species under the summed total $\times$

Total titer is the sum of five measured fatty acids, and summing is a modeling choice in the same sense that the null is. The five species were measured separately, so the choice can be audited rather than argued about.

Every species interacts, and they do not agree on the sign. Recomputing the false-discovery-rate correction within each readout gives 3 to 80 called trigenic interactions per species against 86 on the summed total (Supplementary Fig. 2a). C16:0, C18:0 and C18:1 behave like the total, with 60 to 67 of their calls negative. C14:0 does not: all 26 of its calls are positive and its median trigenic score is +3.90, the largest of any readout. C16:1 is nearly silent, with 2 digenic and 3 trigenic calls, so a claim about it would be a claim about a readout this design cannot resolve.

C14:0 reverses between orders. On C14:0 all 15 digenic calls are negative with a median of -1.40 , and all 26 trigenic calls are positive with a median of $+3.90$ (Supplementary Fig. 2c). The direction of the interaction on that species depends on how many deletions are counted. Three of the five species move down from the digenic to the trigenic median and two move up, so the reversal the main text reports on the summed total is a majority behavior rather than a universal one.

The summed total is not a faithful summary. Correlating τ across readouts over the same 119 triples (Supplementary Fig. 2b) puts the total close to C16:0 and C18:1 at $r = 0.91$, further from C16:1 and C18:0 at $r = 0.54$ and 0.40 , and on the other side of zero from C14:0 at $r = -0.28$. The two most negatively related species are C14:0 and C18:1 at $r = -0.41$. Summing therefore reports the species that carry the most mass and cancels a species whose interactions run the other way, which is the mechanism behind the main text's observation that 31 positive consensus interactions exist across the five species and none survives on their sum.

What follows for a campaign. A campaign that optimizes one species is not optimizing a scaled version of the total, and the interaction structure it would have to navigate is not the one reported here. The reachability result of the main text is stated on total titer, and repeating it per species is a separate analysis this document does not carry out.

References

- [1] Domingo, J., Baeza-Centurion, P. & Lehner, B. The Causes and Consequences of Genetic Interactions (Epistasis). *Annual Review of Genomics and Human Genetics* **20**, 433–460 (2019). URL <https://www.annualreviews.org/content/journals/10.1146/annurev-genom-083118-014857>.
- [2] Costanzo, M. *et al.* A global genetic interaction network maps a wiring diagram of cellular function. *Science* **353**, aaf1420–aaf1420 (2016). URL <https://www.sciencemag.org/lookup/doi/10.1126/science.aaf1420>.
- [3] Baryshnikova, A. *et al.* Quantitative analysis of fitness and genetic interactions in yeast on a genome scale. *Nature Methods* **7**, 1017–1024 (2010). URL <http://www.nature.com/articles/nmeth.1534>.
- [4] Kuzmin, E. *et al.* Systematic analysis of complex genetic interactions. *Science* **360**, eaao1729 (2018). URL <https://www.science.org/doi/10.1126/science.aao1729>.
- [5] Costanzo, M. *et al.* Environmental robustness of the global yeast genetic interaction network. *Science* **372**, eabf8424 (2021). URL <https://www.science.org/doi/full/10.1126/science.abf8424>.
- [6] Johnston, K. E. *et al.* A combinatorially complete epistatic fitness landscape in an enzyme active site. *Proceedings of the National Academy of Sciences* **121**, e2400439121 (2024). URL <https://www.pnas.org/doi/10.1073/pnas.2400439121>.
- [7] Alcantar, M. A., English, M. A., Valeri, J. A. & Collins, J. J. A high-throughput synthetic biology approach for studying combinatorial chromatin-based transcriptional regulation. *Molecular Cell* **84**, 2382–2396.e9 (2024). URL [https://www.cell.com/molecular-cell/abstract/S1097-2765\(24\)00448-9](https://www.cell.com/molecular-cell/abstract/S1097-2765(24)00448-9).
- [8] Goldstein, I., Hale, J. J. & Ehrenreich, I. M. Global epistasis in budding yeast driven by many natural variants whose effects scale with fitness. *Genetics* **231**, iyaf136 (2025). URL <https://doi.org/10.1093/genetics/iyaf136>.
- [9] Matsui, T. *et al.* The interplay of additivity, dominance, and epistasis on fitness in a diploid yeast cross. *Nature Communications* **13**, 1463 (2022). URL <https://www.nature.com/articles/s41467-022-29111-z>.
- [10] Lian, J., Hamedirad, M., Hu, S. & Zhao, H. Combinatorial metabolic engineering using an orthogonal tri-functional CRISPR system. *Nature Communications* **8**, 1688 (2017). URL <http://www.nature.com/articles/s41467-017-01695-x>.
- [11] Kuzmin, E. *et al.* Exploring whole-genome duplicate gene retention with complex genetic interaction analysis. *Science* **368**, eaaz5667 (2020). URL <https://www.science.org/doi/10.1126/science.aaz5667>.
- [12] Volk, M. What the 010 Trigenic Model Is Measured Against: Additive nulls, what each model actually sees, and whether the comparison is fair. Internal document, University of Illinois Urbana-Champaign (2026). Doc Key: notes-tex/010-additive-baselines Git Commit: c7c862632855-dirty Git Branch: main.
- [13] Segrè, D., DeLuna, A., Church, G. M. & Kishony, R. Modular epistasis in yeast metabolism. *Nature Genetics* **37**, 77–83 (2005). URL <https://www.nature.com/articles/ng1489>.
- [14] Kryazhimskiy, S. Emergence and propagation of epistasis in metabolic networks. *eLife* **10**, e60200 (2021). URL <https://doi.org/10.7554/eLife.60200>.
- [15] Sameith, K. *et al.* A high-resolution gene expression atlas of epistasis between gene-specific transcription factors exposes potential mechanisms for genetic interactions. *BMC Biology* **13**, 112 (2015). URL <https://doi.org/10.1186/s12915-015-0222-5>.